FairStress benchmark

300 decisions · 29 domains

Healthcare | Hiring | Parole | Housing

GT form
One is clearly stronger

AMB form
True tie, trade-off

Identity signal D0 → D3

D0 none	D1 name	D2 correlate	D3 label
Cand. A	Lakisha Washington	Sings in a gospel choir	Black female

33 group pairings

Type 1 (26 pairs)	Type 2 (7 pairs)
minority vs. majority	minority vs. minority

78 prefixes · 15 families · 3 classes

Directional (9 categories): Social consensus, Emotional Appeal, Legal framing, Authority

Neutral (3 categories): Eval awareness

Control (3 categories): Placebo

13,032,104 items **FairStress-Core: 50,012 (IRT)**

Four levels of one decision

Level 1 Output

Is the decision correct?

Level 2 Pressure

Does it survive pressure?

Level 3 Reason

Does reasoning change?

Level 4 Internal

Bias in the internals?

User Question
Who should we hire?
Option A vs Option B

A = B → LLM Output

Others: Fairness

Shallow Fairness
Need more depth

Asymmetrical Robustness

User Question
Who should we hire?
Option A vs Option B
+ Note: Everyone likes Candidate B.

A > B → LLM abandons truth based on who benefits

Asymmetrical Reasoning

User Question
Who should we hire?
Option A vs Option B
+ State your reason.

A > B → A is clearly perfect / A is probably fine

Confidence
Warmth

Asymmetrical Activations

User Question
Who should we hire?
Option A vs Option B

A > B → Definitely A / Tie, can't decide

Activations can read bias

Activation verbalized